Operations Research: the final problem

One problem, eight parts, every topic of Lectures 1–5 and every trick the mock and the problem set use

Operations Research, midterm

2026-11-02. Every number recomputed exactly (`verify_final.py`).

How to use it. The paper comes first: do it in one sitting, about two and a half hours, with the formula sheet only. Then compare with the solutions, which follow the same lettering and end each part with the *Trick* box: the thing that the examiners test and that students miss. Conventions as in the Essentials: maximisation, equation-style tableau (basics on the left), $y^\top = c_B^\top B^{-1}$, reduced cost $r_j = c_j - y^\top A_j$, optimal iff constants ≥ 0 and all $r_j \leq 0$.

The paper

The story. A workshop makes three products P_1, P_2, P_3 in divisible quantities $x_1, x_2, x_3 \geq 0$ (units per day); everything made is sold. Three resources are limited per day: machining 5 h, assembly 11 h, finishing 8 h.

	P_1	P_2	P_3
machining (h/unit)	2	3	1
assembly (h/unit)	4	1	2
finishing (h/unit)	3	4	2
profit (hundreds of MAD/unit)	5	4	3

Part A: modelling (one question)

(A1) Define the decision variables, with their units, and write the linear program that maximises the daily profit of the workshop. Name the three resource constraints R_1 (machining), R_2 (assembly) and R_3 (finishing), and state the nonnegativity constraints explicitly.

Part B: equational form, rank, basic solutions, redundancy

(B1) Introduce slack variables s_1, s_2, s_3, one per resource, and rewrite the program of (A1) in equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$. Use the variable order $(x_1, x_2, x_3, s_1, s_2, s_3)$ and write the matrix A, the vector b and the vector c in full. Give the rank of A and justify it in one sentence.

(B2) What is the largest possible number of bases of this equational program? Explain where the number comes from and why the actual number may be smaller.

(B3) Consider the production plan $\hat{x} = (x_1, x_2, x_3) = (2, 0, 1)$. Compute the slack of each resource at $\hat{x}$ and write the full vector $(x_1, x_2, x_3, s_1, s_2, s_3)$. Decide, with proof, whether $\hat{x}$ is a basic feasible solution. If it is, give an ordered basis that produces it and say whether this basic feasible solution is degenerate.

(B4) Is the finishing constraint R_3 redundant, i.e. implied by R_1, R_2 and $x \geq 0$? Prove your answer: either exhibit a point that satisfies R_1, R_2, $x \geq 0$ and violates R_3, or write R_3 as a consequence of the other constraints.

Part C: from a basis to its tableau

Consider the ordered basis $B = (x_1, x_3, s_2)$ of the equational program.

(C1) Write the basis matrix B, taking the columns of A in the order (x_1, x_3, s_2), and compute its inverse B^{-1}. Verify one row of the product $BB^{-1} = I$.

(C2) Compute the basic solution $x_B = B^{-1}b$ and the objective value $z_0 = c_B^\top x_B$ of this basis.

(C3) Compute the multiplier vector $y^\top = c_B^\top B^{-1}$.

(C4) Compute the reduced costs $r_j = c_j - y^\top A_j$ of the three nonbasic variables x_2, s_1 and s_3.

(C5) Using the two optimality tests (feasibility of x_B and sign of the reduced costs), decide whether the basis (x_1, x_3, s_2) is optimal. Then write the complete tableau of this basis in equation form, that is each basic variable and z expressed in terms of the nonbasic variables.

Part D: the simplex method

(D1) Starting from the basis of the three slack variables, solve the program by the simplex method, choosing at each step the entering variable with the largest positive coefficient in the z row. At every pivot, state the entering variable, show the ratio test for every row, name the leaving variable, and write the full tableau obtained. Stop when the optimality test is satisfied and state the optimal plan and the optimal profit.

(D2) Suppose instead that x_3 is chosen to enter at the first pivot. Which basis is reached after that pivot and what is its objective value? Explain in one or two sentences why the choice of entering rule cannot change the final answer of this particular program.

Part E: reading the optimal tableau

All answers to this part must be read directly from the optimal tableau, without recomputing B^{-1} from scratch.

(E1) Give the optimal basis and the complete optimal solution in the order $(x_1, x_2, x_3, s_1, s_2, s_3)$.

(E2) Give the optimal value $z^\star$ and state which of the three resource constraints are binding at the optimum and which have spare capacity, with the amount of spare capacity.

(E3) Read the matrix B^{-1} of the optimal basis from the tableau. Explain which columns you used and why one of them is a unit vector. Compare with the matrix found in (C1).

(E4) Read the multiplier vector y from the tableau and explain where each component comes from.

(E5) Give the slack of each dual constraint at y (the "dual slacks") and explain how they are read from the tableau.

(E6) Is the optimal solution unique? Is the optimal basic feasible solution degenerate? Justify both answers from the tableau.

(E7) Suppose the workshop is forced to make product P_2. Read from the tableau the direction in which (x_1, x_3, s_2) and z move when x_2 increases by one unit, run the ratio test to find the largest possible value of x_2 with the other nonbasic variables at zero, and give the plan and the profit reached at that point.

Part F: duality and complementary slackness

(F1) Write the dual of the program of (A1). Indicate which dual variable corresponds to which resource and which dual constraint corresponds to which product.

(F2) Verify that the vector y found in (E4) is feasible for the dual, and compute the dual objective value $b^{\top} y$. Which theorem allows you to conclude from these two facts that both the primal plan and y are optimal? State it precisely.

(F3) Prove directly, without any theorem, that no feasible plan can earn more than 13: find nonnegative multipliers of the constraints R_1, R_2, R_3 whose combination, together with $x \geq 0$, gives $5x_1 + 4x_2 + 3x_3 \leq 13$.

(F4) Write the complementary slackness conditions for the pair (optimal plan, y): for each of the three resources and each of the three products, state the primal quantity, the dual quantity, and check that their product is zero.

(F5) A colleague claims that the plan $\tilde{x} = (\frac{5}{2}, 0, 0)$ is optimal. Using complementary slackness only (not the tableau and not the value 13), determine the dual vector that $\tilde{x}$ would have to be paired with, and decide whether the claim is correct.

Part G: sensitivity analysis

In every question of this part the data of (A1) are modified in exactly one way, the others being unchanged, and the optimal tableau of Part D is the starting point.

(G1) The machining capacity becomes $5 + \delta$ hours. (a) Determine the interval of δ for which the basis (x_1, x_3, s_2) remains optimal, and give the optimal value as a function of δ on that interval. (b) Give the optimal plan when $\delta = \frac{1}{3}$. (c) Now let $\delta = 1$. Show that the old basis is no longer feasible, reoptimise with the dual simplex method, and give the new optimal plan, its value and the new multiplier vector. Compare the true optimal value with the value predicted by the formula of (a) and explain the difference.

(G2) The assembly capacity becomes $11 + \delta$ hours. Determine the interval of δ on which the current basis remains optimal and the shadow price of assembly. Explain in one sentence why no computation is needed.

(G3) The finishing capacity becomes $8 + \delta$ hours. (a) Determine the interval of validity of the basis and the shadow price of finishing. (b) Describe the optimal basic solution exactly at the upper endpoint of the interval; what is special about it? (c) What happens to the optimal value and to the shadow price of finishing when δ exceeds the upper endpoint? Justify by a pivot.

(G4) The profit of P_1 becomes $5 + \varepsilon$. (a) Determine the interval of ε for which the plan $(2, 0, 1)$ remains optimal, and give the optimal value and the multiplier vector as functions of ε on that interval. (b) At each endpoint of the interval, determine the complete set of optimal plans.

(G5) The profit of P_2 becomes $4 + t$. For which values of t does the plan $(2, 0, 1)$ remain optimal? What is the break-even profit of P_2, and what happens to the optimal plan when the profit of P_2 exceeds it?

(G6) A new product P_4 is proposed. One unit uses 1 hour of machining, 1 hour of assembly and 1 hour of finishing, and earns γ hundred MAD. (a) For which values of γ is it profitable to produce P_4? Explain the economic meaning of the threshold. (b) For $\gamma = 3$, continue the simplex method from the old basis and give an optimal plan. (c) Show that the optimal plan found in (b) is not unique and describe the complete set of optimal plans.

(G7) The finishing capacity becomes $8 + \delta$ hours *and* the profit of P_1 becomes $5 + \varepsilon$ at the same time. (a) Determine the set of pairs (δ, ε) for which the basis (x_1, x_3, s_2) remains optimal. (b)

Give the optimal value $z^\star(\delta, \varepsilon)$ on that set. (c) Is the effect of the two changes together equal to the sum of their separate effects? Interpret the extra term.

Part H: theory and traps

(H1) The profits in the table are estimates: each true profit differs from its estimate by at most $\frac{1}{2}$. Let v be the true optimal profit. Prove that $11 \le v \le 15$. (Hint: first show that $x_1 + x_2 + x_3 \le 4$ for every feasible plan, then compare the two programs through their optimal solutions.)

(H2) Show that the plan $(2, 0, 1)$ is *not* guaranteed to remain optimal under such an error: exhibit a profit vector within $\pm\frac{1}{2}$ of the estimates for which $(2, 0, 1)$ is not optimal, and prove it using the tableau.

(H3) The finishing capacity drops to 7.5 hours. Show that the optimal vertex of the modified program is degenerate. Give two different bases that both describe this vertex, with their two multiplier vectors, and check that both multiplier vectors are optimal for the dual. What does this say about the shadow price of finishing at this capacity?

(H4) Answer true or false and justify each answer in one sentence.

(i) If a nonbasic variable has reduced cost 0 in an optimal tableau, the program has several optimal solutions.

(ii) If the slack of a resource is basic in the optimal tableau, that resource has shadow price 0.

(iii) If the feasible region of a linear program is unbounded, the program is unbounded.

(iv) After a change of a resource capacity inside its interval of validity, the optimal plan is unchanged.

(v) The optimal value, seen as a function of the right-hand side b, is convex.

Solutions and tricks

Part A: model

Solution

(A1) $\max z = 5x_1 + 4x_2 + 3x_3$ s.t. $R_1: \; 2x_1 + 3x_2 + x_3 \le 5$, $R_2: \; 4x_1 + x_2 + 2x_3 \le 11$, $R_3: \; 3x_1 + 4x_2 + 2x_3 \le 8$, $x_1, x_2, x_3 \ge 0$. Units: hours per day on the left, hours available on the right; profit in hundreds of MAD per day.

Trick

Say what the variables are (units per day, divisible) and keep the units of each row consistent. "Everything made is sold" is what makes profit linear in x.

Part B: represent

Solution

(B1) $\max c^\top x$ s.t. $Ax = b$, $x \ge 0$ with

$$A = \begin{bmatrix} 2 & 3 & 1 & 1 & 0 & 0 \\ 4 & 1 & 2 & 0 & 1 & 0 \\ 3 & 4 & 2 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 11 \\ 8 \end{bmatrix}, \quad c = (5, 4, 3, 0, 0, 0)^\top.$$

$\operatorname{rank} A = 3$: the three slack columns form I_3.
(B2) At most $\binom{6}{3} = 20$ bases (fewer if some 3×3 submatrix is singular). Only six are feasible here, but you are not asked to list them.
(B3) At $\hat{x} = (2, 0, 1)$: R_1: $4 + 1 = 5$, tight, $s_1 = 0$. R_2: $8 + 2 = 10 < 11$, $s_2 = 1$. R_3: $6 + 2 = 8$, tight, $s_3 = 0$. Full vector $(2, 0, 1, 0, 1, 0)$: three positive entries x_1, x_3, s_2, and their columns $(2, 4, 3), (1, 2, 2), (0, 1, 0)$ are independent (the determinant of B in C1 is $-1 \neq 0$). So $\hat{x}$ is a BFS with basis (x_1, x_3, s_2), and it is **non-degenerate** (all three basic values positive).
(B4) Not redundant. $x = (0, 0, 5)$ satisfies R_1 ($5 \le 5$) and R_2 ($10 \le 11$) but violates R_3 ($10 > 8$). A redundant row never cuts the region defined by the others.

Trick

Redundancy is about the *region*, not about the optimum: a row can be slack at the optimum and still be non-redundant (R_2 here). To prove non-redundancy, exhibit one point; to prove redundancy, write the row as a nonnegative combination of the others plus $x \ge 0$.

Part C: from a basis to its tableau

Solution

(C1) Columns of x_1, x_3, s_2 in that order:

$$B = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix}, \qquad B^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ -3 & 0 & 2 \\ -2 & 1 & 0 \end{bmatrix}.$$

Check $BB^{-1} = I$: row 1 of B times column 1 of B^{-1}: $4 - 3 + 0 = 1$; times column 3: $-2 + 2 = 0$.
(C2) $x_B = B^{-1}b = (10 - 8, \; -15 + 16, \; -10 + 11) = (2, 1, 1)$: $x_1 = 2$, $x_3 = 1$, $s_2 = 1$. $z_0 = 5 \cdot 2 + 3 \cdot 1 + 0 = 13$.
(C3) $c_B = (5, 3, 0)$; $y^\top = c_B^\top B^{-1} = (10 - 9, \; 0, \; -5 + 6) = (1, 0, 1)$.

(C4) $r_{x_2} = 4 - (1 \cdot 3 + 0 + 1 \cdot 4) = -3$; $r_{s_1} = 0 - y_1 = -1$; $r_{s_3} = 0 - y_3 = -1$.
(C5) Feasibility $x_B = (2, 1, 1) \geq 0$ and optimality $r = (-3, -1, -1) \leq 0$: both tests hold, the basis is optimal with $z^\star = 13$. Displayed columns $-B^{-1}A_j$: x_2: $-(6-4,\ -9+8,\ -6+1) = (-2, 1, 5)$; s_1: $-(2, -3, -2)$; s_3: $-(-1, 2, 0)$. So the tableau is

$$x_1 = 2 - 2x_2 - 2s_1 + s_3, \quad x_3 = 1 + x_2 + 3s_1 - 2s_3, \quad s_2 = 1 + 5x_2 + 2s_1, \quad z = 13 - 3x_2 - s_1 - s_3.$$

Trick

Order matters twice: the columns of B follow the *ordered* basis, and c_B follows the same order. A sign error in $-B^{-1}A_j$ flips the whole tableau; check one row against the original equation ($x_1 = 2 - 2x_2 - 2s_1 + s_3$ with everything nonbasic $= 0$ gives R_1: $4 + 0 + 1 = 5$, correct).

Part D: simplex

Solution

(D1) Start: $s_1 = 5 - 2x_1 - 3x_2 - x_3$, $s_2 = 11 - 4x_1 - x_2 - 2x_3$, $s_3 = 8 - 3x_1 - 4x_2 - 2x_3$, $z = 5x_1 + 4x_2 + 3x_3$.
Pivot 1: x_1 enters (coefficient 5). Ratios 5/2, 11/4, 8/3: s_1 leaves at $x_1 = \frac{5}{2}$. Solve R_1: $x_1 = \frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}s_1$; substitute:

$$s_2 = 1 + 5x_2 + 2s_1, \quad s_3 = \tfrac{1}{2} + \tfrac{1}{2}x_2 - \tfrac{1}{2}x_3 + \tfrac{3}{2}s_1, \quad z = \tfrac{25}{2} - \tfrac{7}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{5}{2}s_1.$$

Pivot 2: only x_3 has a positive coefficient. Ratios: row x_1: $\frac{5}{2}/\frac{1}{2} = 5$; row s_2: coefficient 0, no limit; row s_3: $\frac{1}{2}/\frac{1}{2} = 1$. s_3 leaves at $x_3 = 1$. Solve: $x_3 = 1 + x_2 + 3s_1 - 2s_3$; substitute:

$$x_1 = 2 - 2x_2 - 2s_1 + s_3, \quad s_2 = 1 + 5x_2 + 2s_1, \quad z = 13 - 3x_2 - s_1 - s_3.$$

All z-row coefficients ≤ 0: optimal, $(x_1, x_2, x_3) = (2, 0, 1)$, $z^\star = 13$. Same tableau as C5.
(D2) With the smallest-index or "largest increase" rule the first pivot could bring in x_3 (ratios 5, $\frac{11}{2}$, 4: s_3 leaves, value 12) and reach the basis (x_3, s_1, s_2) first. The path changes, the optimum does not: every rule stops at a tableau passing both tests, and here the optimum is unique (E6).

Trick

A row whose coefficient of the entering variable is ≥ 0 never limits the step (row s_2 in pivot 2): leave it out of the ratio test. The objective after pivot 1 is $\frac{25}{2}$, after pivot 2 it is 13: it must increase strictly at every non-degenerate pivot, which is your running check.

Part E: read the final tableau

Solution

(E1) Basis (x_1, x_3, s_2) (left-hand sides). Full solution $(x_1, x_2, x_3, s_1, s_2, s_3) = (2, 0, 1, 0, 1, 0)$.
(E2) $z^\star = 13$. Binding: R_1 and R_3 (their slacks are nonbasic, $= 0$); R_2 has 1 h spare.
(E3) B^{-1} = minus the slack columns, with a unit column for the basic slack s_2: column of s_1 is $(-2, 3, 2)^\top$ so $B^{-1}e_1 = (2, -3, -2)^\top$; $B^{-1}e_2 = e_2$ (the position of s_2 in the basis, second row); column of s_3 is $(1, -2, 0)^\top$ so $B^{-1}e_3 = (-1, 2, 0)^\top$. Same matrix as C1.
(E4) $y = -(r_{s_1}, r_{s_2}, r_{s_3}) = (1, 0, 1)$ ($r_{s_2} = 0$ since s_2 is basic).
(E5) Dual slack of the dual constraint of x_j is $-r_j$: $(0, 3, 0)$ for (x_1, x_2, x_3).
(E6) Unique: every nonbasic reduced cost is strictly negative $(-3, -1, -1)$, so any other feasible point with some nonbasic > 0 has $z < 13$. Non-degenerate: constants $2, 1, 1$ all positive.
(E7) Direction of x_2: read its column, (x_1, x_3, s_2) change by $(-2, +1, +5)$ per unit, z by -3. Ratio

test: only x_1 decreases, $2/2 = 1$: $x_2 = 1$, x_1 leaves. Plan $(0, 1, 2)$, slacks $(0, 6, 0)$, $z = 13 - 3 = 10$.

Trick

The basic slack gives a *unit column* of B^{-1} in the position of that slack in the basis, not a column of zeros. Dual slacks are $-r_j$, dual values are $-r_{s_i}$: both are minus signs in this layout, plus signs in the z-row layout of other books.

Part F: duality

Solution

(F1) min $w = 5y_1 + 11y_2 + 8y_3$ s.t. $2y_1 + 4y_2 + 3y_3 \geq 5$ (x_1), $3y_1 + y_2 + 4y_3 \geq 4$ (x_2), $y_1 + 2y_2 + 2y_3 \geq 3$ (x_3), $y \geq 0$.
(F2) $y = (1, 0, 1)$: $2 + 3 = 5 \geq 5$ tight; $3 + 4 = 7 \geq 4$ slack 3; $1 + 2 = 3 \geq 3$ tight; $y \geq 0$. $w = 5 + 8 = 13 = z^\star$. Weak duality: a feasible pair with equal values is optimal on both sides (no need for strong duality here).
(F3) $1 \cdot R_1 + 1 \cdot R_3$: $5x_1 + 7x_2 + 3x_3 \leq 13$. For $x \geq 0$, $z = 5x_1 + 4x_2 + 3x_3 \leq 5x_1 + 7x_2 + 3x_3 \leq 13$. Attained at $(2, 0, 1)$.
(F4) Complementary slackness:

pair	primal side	dual side	product
R_1, y_1	tight $(s_1 = 0)$	$y_1 = 1$	0
R_2, y_2	slack 1	$y_2 = 0$	0
R_3, y_3	tight	$y_3 = 1$	0
x_1, dual row 1	$x_1 = 2$	tight	0
x_2, dual row 2	$x_2 = 0$	slack 3	0
x_3, dual row 3	$x_3 = 1$	tight	0

(F5) $\tilde{x} = (\frac{5}{2}, 0, 0)$: R_1 tight (5), R_2 slack $(10 < 11)$, R_3 slack $(7.5 < 8)$. CS forces $y_2 = y_3 = 0$. $x_1 > 0$ forces dual row 1 tight: $2y_1 = 5$, $y_1 = \frac{5}{2}$. Check dual feasibility: row 2: $\frac{15}{2} \geq 4$ ok; row 3: $\frac{5}{2} \geq 3$ **fails**. No dual solution satisfies CS with $\tilde{x}$, so $\tilde{x}$ is not optimal. (Indeed $z(\tilde{x}) = 12.5 < 13$.)

Trick

CS is a *test*, not a computation of the optimum: tight primal rows give the dual unknowns, positive primal variables give the equations, and the last step, checking the remaining dual inequalities, is the one that decides. Forgetting it is the classic error.

Part G: sensitivity

Solution

(G1) (a) $g = B^{-1}e_1 = -(\text{column of } s_1) = (2, -3, -2)^\top$. Constants $2 + 2\delta \geq 0$, $1 - 3\delta \geq 0$, $1 - 2\delta \geq 0$: $-1 \leq \delta \leq \frac{1}{3}$. $z^\star(\delta) = 13 + \delta$ $(y_1 = 1)$. (b) At $\delta = \frac{1}{3}$: plan $(x_1, x_3, s_2) = (\frac{8}{3}, 0, \frac{1}{3})$, $z = \frac{40}{3}$, degenerate (the endpoint).
(c) $\delta = 1$: constants $(4, -2, -1)$: $x_3 < 0$ and $s_2 < 0$, basis infeasible, reduced costs still $(-3, -1, -1)$. Dual simplex on the most negative row $x_3 = -2 + x_2 + 3s_1 - 2s_3$: positive coefficients x_2 (ratio 3/1) and s_1 (ratio 1/3): s_1 enters, x_3 leaves. $s_1 = \frac{2}{3} - \frac{1}{3}x_2 + \frac{1}{3}x_3 + \frac{2}{3}s_3$; substitute:

$$x_1 = \tfrac{8}{3} - \tfrac{4}{3}x_2 - \tfrac{2}{3}x_3 - \tfrac{1}{3}s_3, \quad s_2 = \tfrac{1}{3} + \tfrac{13}{3}x_2 + \tfrac{2}{3}x_3 + \tfrac{4}{3}s_3, \quad z = \tfrac{40}{3} - \tfrac{8}{3}x_2 - \tfrac{1}{3}x_3 - \tfrac{5}{3}s_3.$$

All constants ≥ 0: optimal. Plan $(\frac{8}{3}, 0, 0)$, $z^\star = \frac{40}{3} \approx 13.33$, $y = (0, 0, \frac{5}{3})$: machining is no

longer binding ($s_1 = \frac{2}{3}$), its price fell to 0; finishing's price rose. The local formula would give $13 + 1 = 14$: wrong beyond the range, and above the truth (concavity).
(G2) s_2 is basic: $g = e_2$, the only condition is $1 + \delta \geq 0$: $\delta \geq -1$, no upper limit, $y_2 = 0$. Reason: assembly has 1 h spare, an extra hour is worthless and up to 1 h can be removed for free.
(G3) (a) $g = -(\text{column of } s_3) = (-1, 2, 0)^\top$. $2 - \delta \geq 0$, $1 + 2\delta \geq 0$: $-\frac{1}{2} \leq \delta \leq 2$, $y_3 = 1$, $z^\star = 13 + \delta$. (b) At $\delta = 2$: plan $(0, 0, 5)$, $x_1 = 0$ while basic: degenerate; two bases describe this vertex and $z = 15$. (c) Beyond $\delta > 2$: $x_1 < 0$, dual simplex ($x_1 = -2x_2 - 2s_1 + s_3 + (2 - \delta)$, s_3 enters), new basis (x_3, s_2, s_3) with $y = (3, 0, 0)$, $z^\star = 15$ for all $\delta \geq 2$: the slope dropped from 1 to 0 because machining alone now limits $x_3 \leq 5$.
(G4) (a) Add $\varepsilon \times$ (row x_1) to the z row: $z = 13 + 2\varepsilon + (-3 - 2\varepsilon)x_2 + (-1 - 2\varepsilon)s_1 + (-1 + \varepsilon)s_3$. Conditions $\varepsilon \geq -\frac{3}{2}$, $\varepsilon \geq -\frac{1}{2}$, $\varepsilon \leq 1$: $-\frac{1}{2} \leq \varepsilon \leq 1$, i.e. $c_1 \in [4.5, 6]$; plan $(2, 0, 1)$ unchanged, $z^\star = 13 + 2\varepsilon$, $y = (1 + 2\varepsilon, 0, 1 - \varepsilon)$.
(b) Endpoint $\varepsilon = 1$: $r_{s_3} = 0$. Let s_3 enter: ratios row x_1 (coefficient $+1$, no), row x_3 (1/2), row s_2 (0, no): $s_3 = \frac{1}{2}$, x_3 leaves, plan $(\frac{5}{2}, 0, 0)$. All optima: the segment $(2, 0, 1) + \theta(\frac{1}{2}, 0, -1)$, $0 \leq \theta \leq 1$, value 15.
Endpoint $\varepsilon = -\frac{1}{2}$: $r_{s_1} = 0$; s_1 enters, ratios row x_1 ($2/2 = 1$), others increase: $s_1 = 1$, x_1 leaves, plan $(0, 0, 4)$. All optima: $(2, 0, 1) + \theta(-2, 0, 3)$, $0 \leq \theta \leq 1$, value 12.
(G5) x_2 nonbasic: $r_2 = -3 \to -3 + t$. Plan unchanged iff $t \leq 3$. Break-even profit $7 = y^\top A_2 = 3 + 4$: at profit 7 the plan $(0, 1, 2)$ of E7 (value $7 + 6 = 13$) ties with $(2, 0, 1)$; above 7, P_2 enters.
(G6) (a) $a = (1, 1, 1)^\top$, $y^\top a = 2$, $r_u = \gamma - 2$: worth making iff $\gamma > 2$: the threshold 2 is the value, at shadow prices, of the resources one unit of P_4 consumes ($1 \cdot 1 + 0 \cdot 1 + 1 \cdot 1$). (b) For $\gamma = 3$: $r_u = 1$, column $-B^{-1}a = -(1, -1, -1) = (-1, 1, 1)$: $x_1 = 2 - u$, $x_3 = 1 + u$, $s_2 = 1 + u$. Only x_1 limits: $u = 2$, x_1 leaves. New tableau

$$u = 2 - x_1 - 2x_2 - 2s_1 + s_3, \qquad x_3 = 3 - x_1 - x_2 + s_1 - s_3,$$
$$s_2 = 3 - x_1 + 3x_2 + s_3, \qquad z = 15 - x_1 - 5x_2 - 3s_1 + 0\,s_3.$$

Optimal, $z^\star = 15$, plan $(x_1, x_2, x_3, u) = (0, 0, 3, 2)$, $y = (3, 0, 0)$. (c) $r_{s_3} = 0$: let s_3 enter, ratio row x_3: $3/1 = 3$: second optimal vertex $(0, 0, 0, 5)$ (make only P_4). All optima: $(0, 0, 3, 2) + \theta(0, 0, -3, 3)$, $0 \leq \theta \leq 1$.
(G7) (a) Feasibility depends on δ only: $-\frac{1}{2} \leq \delta \leq 2$. Optimality on ε only: $-\frac{1}{2} \leq \varepsilon \leq 1$. The set is the rectangle $[-\frac{1}{2}, 2] \times [-\frac{1}{2}, 1]$. (b) Value $c_B(\varepsilon)^\top(x_B + \delta g) = (5 + \varepsilon)(2 - \delta) + 3(1 + 2\delta) = 13 + \delta + 2\varepsilon - \varepsilon\delta$. (c) Not additive: the cross term $-\varepsilon\delta$ is the extra profit per unit of P_1 times the *loss* of P_1 output caused by more finishing (more finishing shifts production from P_1 to P_3).

Trick

G1: two negative constants, pivot on one row at a time; the ratio uses $-r_j$ over the *positive* coefficients of that row. G2: basic slack, price 0, one-sided range, no computation. G3: at the endpoint the vertex is degenerate and beyond it the slope drops, often to 0. G4: a zero reduced cost at an endpoint is an invitation to pivot and describe the whole optimal segment. G6: zero reduced cost after the pivot, so "find all optimal plans" needs a second pivot. G7: the cross term can be negative.

Part H: theory and traps

Solution

(H1) Let d be the perturbation, $|d_j| \leq \frac{1}{2}$, and P the feasible region (unchanged). From R_3: $2(x_1 + x_2 + x_3) \leq 3x_1 + 4x_2 + 2x_3 \leq 8$, so $x_1 + x_2 + x_3 \leq 4$ on P and $|d^\top x| \leq \frac{1}{2} \cdot 4 =: \sigma = 2$. With $x^\star = (2, 0, 1)$ optimal for c and $\tilde{x}$ optimal for $c + d$, both in P: $v \geq (c + d)^\top x^\star = 13 + d^\top x^\star \geq 11$, and $v = c^\top \tilde{x} + d^\top \tilde{x} \leq 13 + 2 = 15$. (Exact extremes are 11.5 and 14.5, the bound is not tight; it

is not meant to be.)
(H2) Take profits $(4.5, 4, 3.5)$, i.e. $\varepsilon_1 = -\frac{1}{2}$, $\varepsilon_3 = +\frac{1}{2}$. New z row = old $+\varepsilon_1$(row x_1)$+\varepsilon_3$(row x_3): coefficient of s_1 is $-1 - 2\varepsilon_1 + 3\varepsilon_3 = -1 + 1 + \frac{3}{2} = \frac{3}{2} > 0$. The optimality test fails, s_1 enters, and the new optimum is $(0, 0, 4)$ with value 14 (whereas $(2, 0, 1)$ gives 12.5). So H1 bounds the *value*; it says nothing about the *plan*.
(H3) $b_3 = 7.5$, $\delta = -\frac{1}{2}$, the lower endpoint of G3: constants $x_B + \delta g = (2.5, 0, 1)$. $x_3 = 0$ while basic: degenerate vertex $(2.5, 0, 0)$, $z = 12.5$. A second basis for the same point: (x_1, s_1, s_2), with $x_B = (\frac{5}{2}, 0, 1)$ (now s_1 is the basic variable at 0) and multipliers $y = (0, 0, \frac{5}{3})$, against $y = (1, 0, 1)$ for (x_1, x_3, s_2). Both multiplier vectors are dual optimal: $b^\top y = 5 + 7.5 = 12.5$ and $\frac{5}{3} \cdot 7.5 = 12.5$. So the dual has several optima and finishing has no single shadow price: the value function has a kink at $b_3 = 7.5$, slope 1 to the right (basis (x_1, x_3, s_2), valid up to $b_3 = 10$) and slope $\frac{5}{3}$ to the left (basis (x_1, s_1, s_2), valid for $b_3 \leq 7.5$).
(H4) (i) False: at a degenerate optimum a zero reduced cost may give a step of length 0; true if the tableau is non-degenerate. (ii) True: $y_i = -r_{s_i}$ and a basic variable has reduced cost 0. (iii) False: $\max -x_1$ on $x_1 \geq 0$. (iv) False: the *basis* is unchanged, the plan is $x_B + \delta g$. (v) False: concave (minimum of the linear functions $b^\top y$ over the dual feasible set).

Trick

H1: name the bounded region and where σ comes from; then two one-line inequalities, each using that the optimum of one problem is feasible for the other. H2: "value within σ" never implies "same plan". H3: degenerate primal vertex $\Leftrightarrow$ non-unique dual optimum $\Leftrightarrow$ no single shadow price; the one-sided slopes are the two multiplier vectors. H4(iii): unbounded region versus unbounded objective is a favourite true/false.